

Manolis Savva

Associate Professor & Canada Research Chair (Tier II) in Computer Graphics

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1 Education

2016 Ph.D. in Computer Science, Department of Computer Science, Stanford University

Thesis: Body-centric Understanding of 3D Environments, *Advisor:* Pat Hanrahan

2012 M.Sc. in Computer Science, Stanford University

2009 B.A. in Physics and Computer Science, Cornell University

2 Employment History

09/2024 - current Associate Professor, School of Computing Science, Simon Fraser University

01/2019 - 08/2024 Assistant Professor, School of Computing Science, Simon Fraser University

07/2018 - 08/2020 Visiting Researcher, Facebook AI Research, Menlo Park, CA

09/2017 - 07/2018 Research Engineer Contractor, AutoRoboto LLC, on-site at Google, Mountain View, CA

09/2017 - 09/2018 Visiting Research Collaborator, Princeton University, Princeton, NJ

08/2016 - 08/2017 Postdoctoral Research Associate, Princeton University, Princeton, NJ

09/2013 - 12/2013 Research Intern, Advanced Technologies Division, Square Enix Co., Ltd., Tokyo, Japan.

05/2009 - 05/2010 Research Assistant, Program of Computer Graphics, Cornell University, Ithaca, NY.

08/2007 - 04/2009 Research Assistant, Lab of Atomic and Solid State Physics, Cornell University, Ithaca, NY.

3 Research Interests Summary

My research focuses on contributions at the intersection of computer graphics, 3D computer vision, and AI. It is characterized by four research themes: human-centric 3D scene representations, large-scale semantically annotated 3D datasets, 3D scene understanding, and 3D simulation for AI. These themes help to reinforce each other—i.e., they tackle the representations, data, algorithms, and systems aspects and are all aimed at enabling new research directions. The overall goal is to provide catalysts for future work in embodied AI, 3D computer vision, robotics, and other areas concerned with modeling, understanding, and generating digital representations of people, places, and things. Downstream applications in a variety of domains stand to benefit from this work. Architectural visualization, visual effects production for films, content creation for games and VR/AR experiences, ergonomics analysis, assistive robotics, and context-aware digital assistants are a few examples. My work has been cited 35,000 times with h-index 52 (Google Scholar Apr 2025).

4 Awards and Recognitions

2025 Canada Research Chair (Tier 2, Renewal) in Computer Graphics

2023 ICLR Outstanding Paper Award, for “Emergence of Maps” paper (4 awards / 4900 submissions)

2022 CHCCS/SCDHM Graphics Interface Early Career Researcher Award

2020 Canada Research Chair (Tier 2) in Computer Graphics

2020 SGP 2020 Dataset Award, for ScanNet dataset

2019 ICCV Best Paper Award Nominee, for Habitat paper (11 selected / 4303 submissions)

2018 SGP 2018 Dataset Award, for ShapeNet dataset

5 Leadership and Service Roles Summary

Departmental & University Committees: SFU Visual & Interactive Computing Institute (VINCI) co-director 2024-; SFU AI Strategy Research Task Force 2024-; SFU CS Graduate Programs Chair 2024-; SFU CS Director Search Committee 2024; SFU CS Faculty Search Committee 2019-20; SFU CS Diversity Committee 2019-20.

Conferences: 3DV 2026 General Chair; Graphics Interface 2021 Program Chair; SIGGRAPH Exec Nominations Committee 2019-2022; SIGGRAPH 2025 Program Committee Paper Sorting; SIGGRAPH 2023, SIGGRAPH Asia 2024 Program Committees; SIGGRAPH Asia 2024 Courses Program Committee; EuroGraphics 2019 Program Committee; Graphics Interface 2019-20 Program Committee; Senior AC for CVPR 2026; AC for CVPR 2020-21,23-25; AC for ECCV 2024; AC for NeurIPS 2023-25; AC for ICCV 2023,2025.

Workshops: AI for 3D Generation at CVPR 2024; Learning to Generate 3D Shapes and Scenes at CVPR 2021, ECCV 2022; ScanNet Benchmark at CVPR 2019-23; Learning 3D Generative Models at CVPR 2020; Embodied AI Workshop at CVPR 2020-24; Habitat Embodied Agents and Challenge at CVPR 2019; 3D Scene Generation at CVPR 2019; Visual Learning and Embodied Agents in Simulation Environments at ECCV 2018; Learning to See from 3D Data at ICCV 2017; Large-scale 3D Shape Retrieval SHREC Track at Eurographics 3DOR 2016-2017.

Conference & Journal Reviewer: ICLR 2020, 3DV 2020-22; ACM CHI 2014,17,20; NeurIPS 2019-22; CVPR 2018,22; ICCV 2021; Eurographics 2018,20; ACM SIGGRAPH 2013,15-18,24; ACM SIGGRAPH Asia 2012,16-18,23; ACM TOG 2017-23; IEEE TPAMI 2022; IEEE TVCG 2013,16-22; CoRL 2017; ACM UIST 2014,16-17.

6 Select Representative Publications

For a complete list of publications please see my Google Scholar Profile. **Legend:** * equal contribution. Names in **bold** are my students. [C] Refereed conference papers, [J] Refereed journal papers, [N]: Non-refereed publications.

- [C9] **W. Peng**, J. Lv, C. Wu, M. Savva. iTACO: Interactable Digital Twins of Articulated Objects from Casually Captured RGBD Videos. *3DV* 2026.
- [C8] **J. Liu**, D. Iliash, A. X. Chang, M. Savva, A. Mahdavi-Amiri. SINGAPO: Single Image Controlled Generation of Articulated Parts in Objects. *ICLR* 2025.
- [C7] **H. Tam**, **H. Pun**, A. Wang, A. X. Chang, M. Savva. SceneMotifCoder: Example-driven Visual Program Learning for Generating 3D Object Arrangements. *3DV* 2025 (oral).
- [C6] **J. Liu**, **H. Tam**, A. Mahdavi-Amiri, M. Savva. CAGE: Controllable Articulation GEneration. *CVPR* 2024.
- [C5] M. Khanna*, **Y. Mao***, H. Jiang, **S. Haresh**, B. Schacklett, D. Batra, A. Clegg, E. Undersander, A. X. Chang, M. Savva. Habitat Synthetic Scenes Dataset (HSSD-200): An Analysis of 3D Scene Scale and Realism Tradeoffs for ObjectGoal Navigation. *CVPR* 2024.
- [C4] K. Grauman, ... , **S. Haresh**, ... , **Y. Mao**, ... , M. Savva, J. Shi, M. Z. Shou, M. Wray. Ego-Exo4D: Understanding Skilled Human Activity from First- and Third-Person Perspectives. *CVPR* 2024.
- [C3] **J. Liu**, A. Mahdavi-Amiri, M. Savva. PARIS: Part-level Reconstruction and Motion Analysis for Articulated Objects. *ICCV* 2023.
- [C2] **Y. Mao**, Y. Zhang, H. Jiang, A. X. Chang, M. Savva. MultiScan: Scalable RGBD scanning for 3D environments with articulated objects. *NeurIPS* 2022.
- [C1] M. Savva*, A. Kadian*, O. Maksymets*, Y. Zhao, E. Wijmans, B. Jain, J. Straub, J. Liu, V. Koltun, J. Malik, D. Parikh, D. Batra. Habitat: A Platform for Embodied AI Research. *ICCV 2019, Best Paper Award Nomination*
- [J2] **J. Liu**, M. Savva, A. Mahdavi-Amiri. Survey on Modeling of Human-made Articulated Objects. *Eurographics STAR, Computer Graphics Forum* 2025
- [J1] A. Kadian*, J. Truong*, A. Gokaslan, A. Clegg, E. Wijmans, S. Lee, M. Savva, S. Chernova, D. Batra. Are We Making Real Progress in Simulated Environments? Measuring the Sim2Real Gap in Embodied Visual Navigation. *Robotics and Automation Letters (RA-L) & IROS* 2020
- [N2] D. Batra, A. X. Chang, S. Chernova, A. J. Davison, J. Deng, V. Koltun, S. Levine, J. Malik, I. Mordatch, R. Mottaghi, M. Savva, H. Su. Rearrangement: A Challenge for Embodied AI. *arXiv:2011.01975 [cs.AI]*, 2020
- [N1] A. X. Chang, T. Funkhouser, L. Guibas, P. Hanrahan, Q. Huang, Z. Li, S. Savarese, M. Savva, S. Song, H. Su, J. Xiao, L. Yi, and F. Yu. ShapeNet: An Information-Rich 3D Model Repository. *arXiv:1512.03012*, 2015